Sectional, assembly & exploded views

Higher · Drawing & spatial awareness · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State the name of two types of section other than a full section, and describe when each is used.

- Part section: a small area cut away to show one internal feature
- Revolved section: shows the cross-section of an arm or spoke rotated on the view
- Stepped section: the cutting plane changes direction to pass through several features
- Half section: shows inside and outside of a symmetrical part
- Removed section: a revolved section drawn away from the view

3 Quick check

- a) Parts are aligned along assembly lines
- b) A bolt
- c) Revolved section
- d) 45°

4 Key terms

- 1 – C
- 2 – B
- 3 – E
- 4 – A
- 5 – D

5 Explain which parts should be hatched and which should not.

- The casing (cut material) is hatched with thin lines at 45°
- The shaft, nut and bolt are not hatched when cut lengthways
- The web is not hatched when cut along its length
- Adjacent parts are hatched in different directions or spacings

6 Explain what is meant by ‘Stepped section’.

- The cutting plane changes direction to cut through several features.

7 Explain what is meant by ‘Part section’.

- A small cut-away area, bounded by a wavy line, to show one feature.